F FWallSim · Firewall Course

CLASS 12

SD-WAN with two ISPs

SD-WAN members, performance SLA, failover

Advanced | ~30 minutes | FWallSim v1.0.0 (FortiOS 7.4 style)

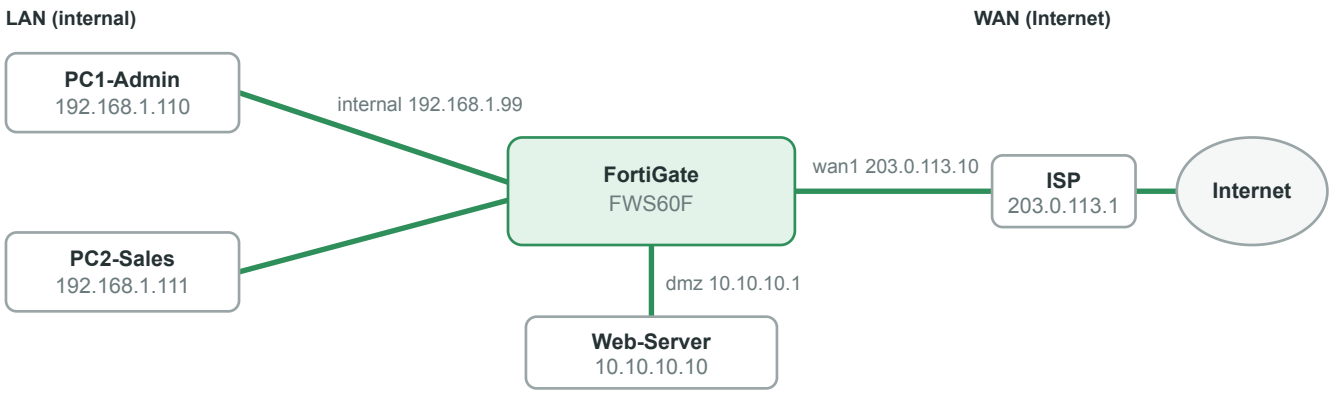
YOU WILL LEARN

- SD-WAN members
- performance SLA
- failover

BEFORE YOU START

Finish Class 11 first.

The lab network



The goal

THE IDEA

A second internet line (ISP2) is now plugged into wan2. Build SD-WAN so the office survives an ISP1 outage automatically.

How to start

1. Open FWallSim and log in.
2. Click **Labs** in the header.
3. Find **SD-WAN with two ISPs** and click **Start**.
4. The Lab Guide panel opens. Each task below gets a green tick as soon as it works.

Tasks

1 SD-WAN is enabled with wan1 and wan2 as members

Network > SD-WAN > SD-WAN Zones: enable, add wan1 (gw 203.0.113.1) and wan2 (gw 198.18.10.1). FortiOS refuses while policies still use wan1 directly: first change policies 1 and 3 (Outgoing) and 2 (Incoming) to virtual-wan-link. Do that right after enabling SD-WAN with wan2 only, then add wan1.

2 Policy 1 goes out the SD-WAN zone and a default route points to it

Edit policy 1: Outgoing interface virtual-wan-link. Static Routes: 0.0.0.0/0, Interface = SD-WAN zone virtual-wan-link.

3 A Performance SLA probes an internet server (not the ISP gateway)

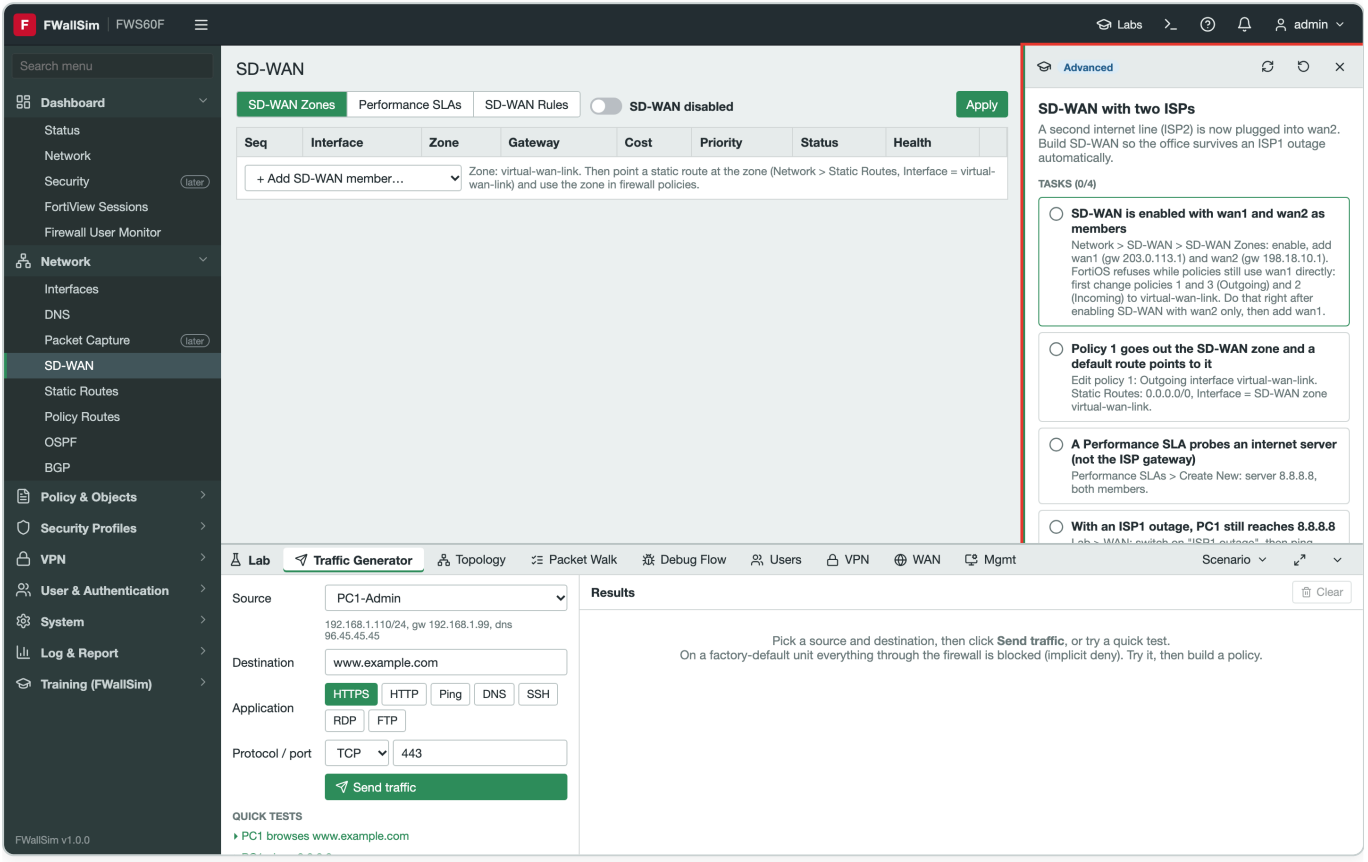
Performance SLAs > Create New: server 8.8.8.8, both members.

4 With an ISP1 outage, PC1 still reaches 8.8.8.8

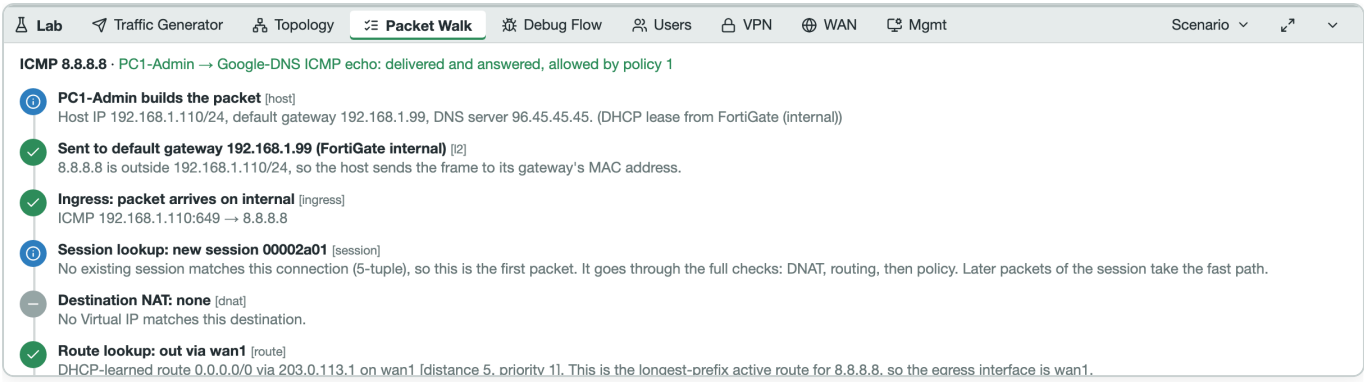
Lab > WAN: switch on "ISP1 outage", then ping 8.8.8.8 from PC1.

Screenshots

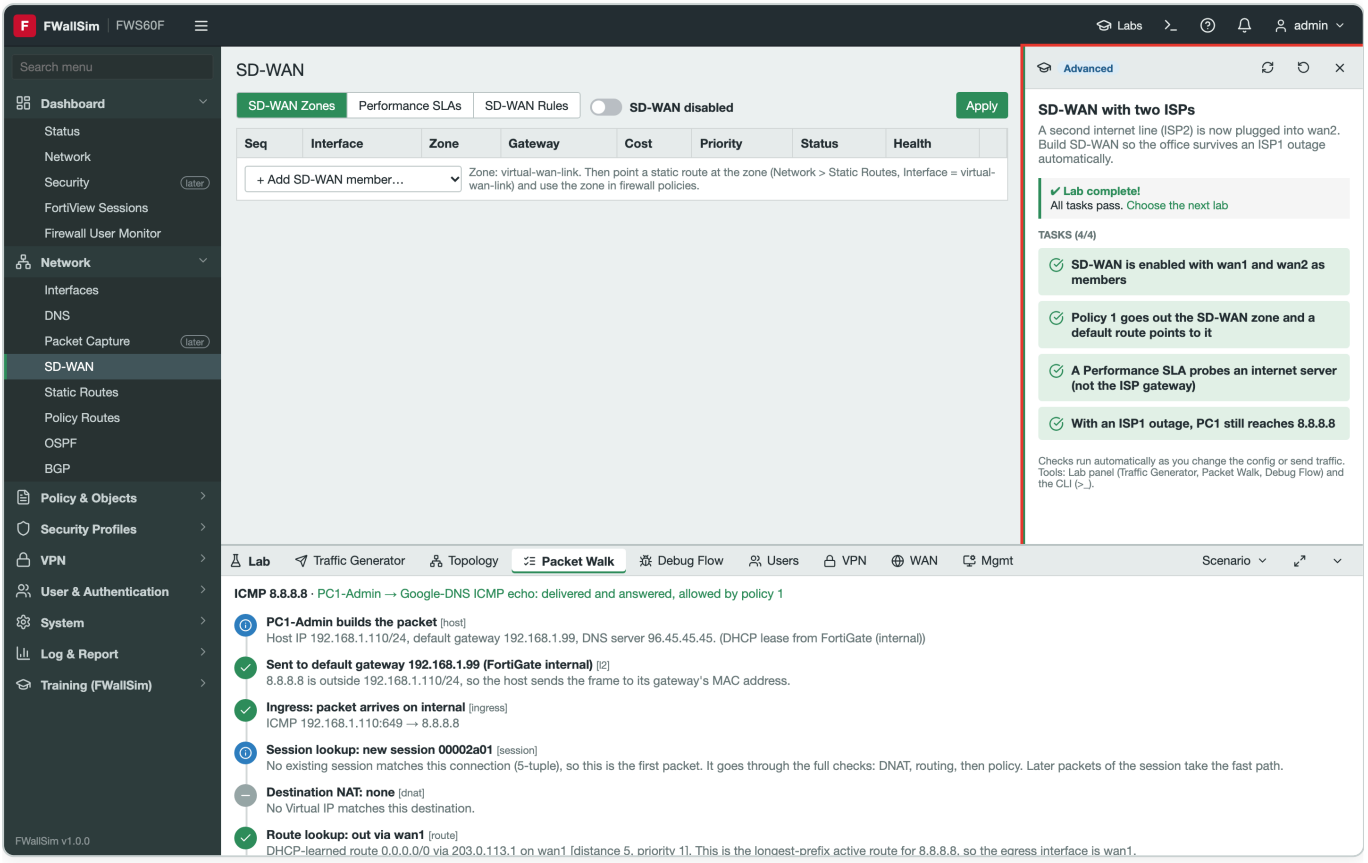
Numbered red markers show where to click or what to look at.



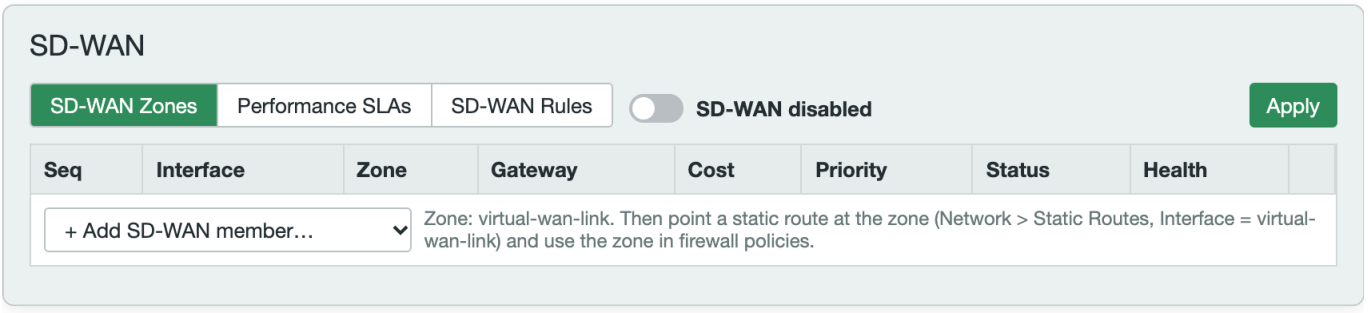
Lab Started



Packet Walk Before



Lab Solved



SD-WAN

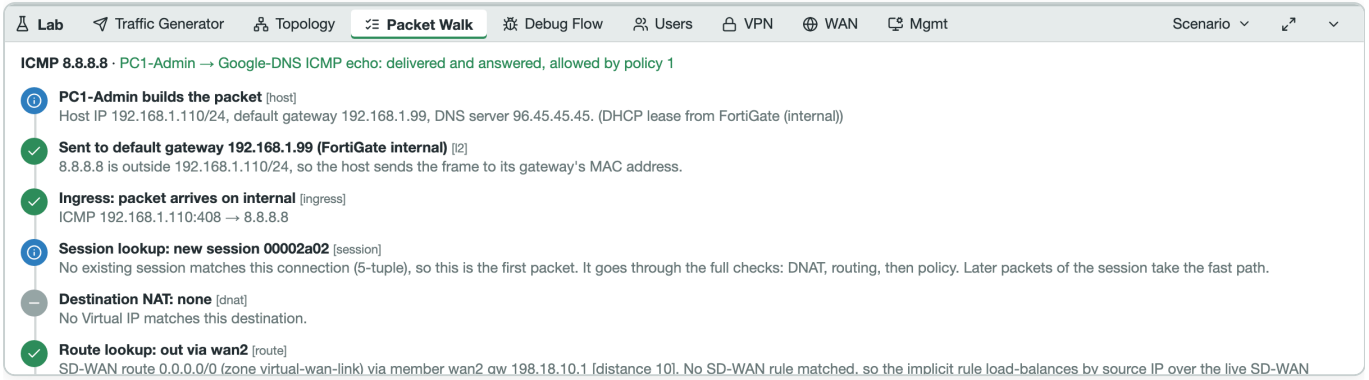
Static Routes

+ Create New | Edit | Clone | Delete | Search

Destination	Gateway IP	Interface	Distance	Priority	Status	Comments
0.0.0.0/0	0.0.0.0	SD-WAN zone virtual-wan-link	10	1	Active	

The default route learned by DHCP on wan1 does not appear here. See it under Dashboard > Network > Routing. Lower distance wins for the same prefix; for equal distance, lower priority wins.

Static Routes



Packet Walk After

Next: Class 13: Dynamic routing: OSPF over VPN and BGP